

MDS5202 Assignment 1

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Conceptual

1. For each of parts (a) through (d), indicate whether we would generally expect the performance of a flexible statistical learning method to be better or worse than an inflexible method. Justify your answer.
 - (a) The sample size n is extremely large, and the number of predictors p is small.
 - (b) The number of predictors p is extremely large, and the number of observations n is small.
 - (c) The relationship between the predictors and response is highly non-linear.
 - (d) The variance of the error terms, i.e. $\sigma^2 = \text{Var}(\epsilon)$, is extremely high.
3. We now revisit the bias-variance decomposition.
 - (a) Provide a sketch of typical (squared) bias, variance, training error, test error, and Bayes (or irreducible) error curves, on a single plot, as we go from less flexible statistical learning methods towards more flexible approaches. The x-axis should represent the amount of flexibility in the method, and the y-axis should represent the values for each curve. There should be five curves. Make sure to label each one.
 - (b) Explain why each of the five curves has the shape displayed in part (a).

Applied

8. This exercise relates to the **College** data set, which can be found in the file **College.csv** in the additional file. It contains a number of variables for 777 different universities and colleges in the US. The variables are:
 - **Private**: Public/private indicator
 - **Apps**: Number of applications received
 - **Accept**: Number of applicants accepted
 - **Enroll**: Number of new students enrolled
 - **Top10perc**: New students from top 10% of high school class
 - **Top25perc**: New students from top 25% of high school class
 - **F.Undergrad**: Number of full-time undergraduates

- **P.Undergrad**: Number of part-time undergraduates
- **Outstate**: Out-of-state tuition
- **Room.Board**: Room and board costs
- **Books**: Estimated book costs
- **Personal**: Estimated personal spending
- **PhD**: Percent of faculty with Ph.D.'s
- **Terminal**: Percent of faculty with terminal degree
- **S.F.Ratio**: Student/faculty ratio
- **perc.alumni**: Percent of alumni who donate
- **Expend**: Instructional expenditure per student
- **Grad.Rate**: Graduation rate

Before reading the data into R, it can be viewed in Excel or a text editor.

- Use the `read.csv()` function to read the data into R. Call the loaded data `college`. Make sure that you have the directory set to the correct location for the data.
- Look at the data using the `View()` function. You should notice that the first column is just the name of each university. We don't really want R to treat this as data. However, it may be handy to have these names for later. Try the following commands:

```
> rownames(college) <- college[, 1]
> View(college)
```

You should see that there is now a `row.names` column with the name of each university recorded. This means that R has given each row a name corresponding to the appropriate university. R will not try to perform calculations on the row names. However, we still need to eliminate the first column in the data where the names are stored. Try

```
> college <- college[, -1]
> View(college)
```

Now you should see that the first data column is **Private**. Note that another column labeled `row.names` now appears before the **Private** column. However, this is not a data column but rather the name that R is giving to each row.

- Use the `summary()` function to produce a numerical summary of the variables in the data set.
 - Use the `pairs()` function to produce a scatterplot matrix of the first ten columns or variables of the data. Recall that you can reference the first ten columns of a matrix `A` using `A[,1:10]`.
 - Use the `plot()` function to produce side-by-side boxplots of **Outstate** versus **Private**.
 - Create a new qualitative variable, called **Elite**, by binning the **Top10perc** variable. We are going to divide universities into two groups based on whether or not the proportion of students coming from the top 10% of their high school classes exceeds 50%.

```

> Elite <- rep("No", nrow(college))
> Elite[college$Top10perc > 50] <- "Yes"
> Elite <- as.factor(Elite)
> college <- data.frame(college, Elite)

```

Use the `summary()` function to see how many elite universities there are. Now use the `plot()` function to produce side-by-side boxplots of `Outstate` versus `Elite`.

- v. Use the `hist()` function to produce some histograms with differing numbers of bins for a few of the quantitative variables. You may find the command `par(mfrow = c(2, 2))` useful: it will divide the print window into four regions so that four plots can be made simultaneously. Modifying the arguments to this function will divide the screen in other ways.
- vi. Continue exploring the data, and provide a brief summary of what you discover.

10. This exercise involves the `Boston` housing data set.

- (a) To begin, load in the `Boston` data set. The `Boston` data set is part of the `ISLR2` library.

```

> library(ISLR2)

```

Now the data set is contained in the object `Boston`.

```

> Boston

```

Read about the data set:

```

> ?Boston

```

How many rows are in this data set? How many columns? What do the rows and columns represent?

- (b) Make some pairwise scatterplots of the predictors (columns) in this data set. Describe your findings.
- (c) Are any of the predictors associated with per capita crime rate? If so, explain the relationship.
- (d) Do any of the census tracts of Boston appear to have particularly high crime rates? Tax rates? Pupil-teacher ratios? Comment on the range of each predictor.
- (e) How many of the census tracts in this data set bound the Charles river?
- (f) What is the median pupil-teacher ratio among the towns in this data set?
- (g) Which census tract of Boston has lowest median value of owner-occupied homes? What are the values of the other predictors for that census tract, and how do those values compare to the overall ranges for those predictors? Comment on your findings.
- (h) In this data set, how many of the census tracts average more than seven rooms per dwelling? More than eight rooms per dwelling? Comment on the census tracts that average more than eight rooms per dwelling.